

DSAII- HW1

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Problem 1

$o(n)$ says for $o(g(n)) = f(n)$ for all $c > 0$ there exists $n_0 > 0$ such that $0 \leq f(n) < cg(n)$ for all $n \geq n_0$

Part A

For 2 phones we want to minimize the worst case total drops, as implied by $o(n)$ from above. Suppose we drop the first phone at intervals of size k , so the drops for the first phone are $k, 2k, 3k \dots$

The worst case for the 1st phone drop is n/k , where n is the number of rungs we dropped from. The Worst Case for the 2nd Phone would be $k - 1$ drops, since we would have to drop it from every rung within the interval k

Thus the total worse case is $(n/k) + k - 1$

To minimize, we can take the derivative and solve for 0.

$$d/dk[(n/k) + k - 1] = (-n/k^2) + 1$$

$$0 = (-n/k^2) + 1$$

$$nk^2 = n$$

$$k = \sqrt{n}$$

This Algorithm can be characterized in 2 setps:

1. Drop the phone $\sqrt{n}, 2\sqrt{n}, 3\sqrt{n} \dots$ rungs until the 1st phone breaks. We call this rung $r\sqrt{n}$.
2. Then Linearly drop the 2nd phone at rungs $(r - 1)\sqrt{n} + 1$ to $r\sqrt{n} - 1$ until we find which rung it breaks at.

The run time for the algorithm is $\sqrt{n}$ for the 1st phone and at most $\sqrt{n} - 1$ for the 2nd phone which is just $O(\sqrt{n})$, thus $o(n)$ total.

Part B

Let $f_k(n)$ be the minimum number of drops n for k phones

Using the same Structure from *Part A*. For the Base Case, we have a linear search for the 1st phone, $f_1(n) = n$ and for all subsequent phones we have only have to do only only 1 drop $f_k(1) = 1$.

With only 2 phones we got $O(\sqrt{n})$ by using the intervals. Since we have k phones we can find a patterns using the same idea.

For 3 phones, $k = 3$, the interval becomes $n^{1/3}$.

We now have 2 phones, and $n^{2/3}$ rungs, so $f_2(n^{2/3}) = O(n^{1/3})$

In total, $O(n^{1/3}) = n^{1/3} + n^{1/3}$

The General Patterns for k phones is that we have the interval of n rungs, n^1 , with each interval having $n^{k-1/k}$ rungs.

The 1st phone is used at most $n^{1/k}$ times to find the interval.

We can recursively use the $k - 1$ phones on the $n^{k-1/k}$ rungs.

This recurrence relation is $f_k(n) = n^{1/k} + f_{k-1}(n^{k-1/k})$

This recurrence has $O(n^{1/k})$ runtime which we can verify

$$f_k(n) = n^1 + f_{k-1}(n^{k-1/k})$$

$$f_k(n) = n^1 + O(n^{k-1/k})^{1/(k-1)}$$

$$f_k(n) = n^1 + O(n^{1/k})$$

$$f_k(n) = O(n^{1/k})$$

Thus $n^{1/k} = o(n^{1-1/k})$ since $1/k < 1/(k-1)$

Problem 2

Part A

Given the Array A of size n , the probability of any particular element chosen is $1/n$.

Let X_i be an Indicator Random Variable where

$X_i = 1$ if the i^{th} smallest element is chosen as the pivot, 0 otherwise.

Since there are n elements the Expected value for X_i is

$$E[X_i] = P(X_i = 1) = 1/n \text{ since } \sum_{i=1}^n X_i = 1$$

Part B

Let $T(n)$ be a Random Variable for the Runtime of the Quick Sort on Array A with n elements. We must show that

$$E[T(n)] = E[\sum_{q=1}^n X_q(T(q-1) + T(n-q) + \Theta(n))]$$

For QS, we select a pivot at random. If the pivot is the Q^{th} smallest element, then $X_q = 1$.

This fact implies that the Left Array A_L has $q - 1$ elements and takes $T(q - 1)$ time and the Right Array A_R has $n - q$ elements and takes $T(n - q)$ time. Partitioning in the both the worst case and the best case takes n times, which is $\Theta(n)$ runtime.

For any element q , $X_q = 1$, thus we can directly apply the expectation and get

$$E[T(n)] = E[\sum_{q=1}^n X_q(T(q-1) + T(n-q) + \Theta(n))]$$

Part C

To simplify the solution from Part B, we can use the Expectation from Part A, $E[X_q] = 1/n$ and the LOE (which was applied in Part B).

$$E[T(n)] = E[\sum_{q=1}^n X_q(T(q-1) + T(n-q) + \Theta(n))]$$

$$E[T(n)] = \sum_{q=1}^n E[X_q] \cdot E[T(q-1) + T(n-q) + \Theta(n)]$$

$$E[T(n)] = \sum_{q=1}^n 1/n \cdot (E[T(q-1)] + E[T(n-q)] + \Theta(n))$$

$$E[T(n)] = 1/n \sum_{q=1}^n E[T(q-1)] + 1/n \sum_{q=1}^n E[T(n-q)] + 1/n \sum_{q=1}^n \Theta(n)$$

$$E[T(n)] = 2/n \sum_{q=1}^{n-1} E[T(q)] + \Theta(n)$$

Note that the summation can start at 1 since there is no $n = 0$ case and when $n = 1$ the summation is from 1 to $n - 1$ elements since the 1st is selected as the pivot.

Part D

We need to show the bound of the Summation for this part.

$$\sum_{q=1}^{n-1} q \lg q \leq n^2/2 \lg n - n^2/8 \text{ for } n \geq 2$$

Splitting the sum gives us

$$\sum_{q=1}^{n-1} q \lg q = \sum_{q=1}^{\lfloor n/2 \rfloor} q \lg q + \sum_{q=\lfloor n/2 \rfloor}^{n-1} q \lg q$$

For the 1st sum, $q < n/2$ implies $q < \lg(n/2) = \lg n - 1$

$$\sum_{q=1}^{\lfloor n/2 \rfloor} q \lg q \leq \sum_{q=1}^{\lfloor n/2 \rfloor} q(\lg n - 1)$$

For the 2nd sum $q \leq n - 1$ implies $\lg q \leq \lg n$

$$\sum_{q=\lfloor n/2 \rfloor}^{n-1} q \lg q \leq \sum_{q=\lfloor n/2 \rfloor}^{n-1} q \lg n$$

Combining the 2 we get

$$\sum_{q=1}^{n-1} q \lg q \leq (\lg n - 1) \sum_{q=1}^{\lfloor n/2 \rfloor} q + \lg n \sum_{q=\lfloor n/2 \rfloor}^{n-1} q$$

$$\sum_{q=1}^{n-1} q \lg q \leq \lg n \sum_{q=1}^{n-1} q - \sum_{q=1}^{\lfloor n/2 \rfloor} q$$

$$\sum_{q=1}^{n-1} q \lg q \leq \lg n \frac{(n-1)n}{2} - \frac{(\lfloor n/2 \rfloor - 1)\lfloor n/2 \rfloor}{2}$$

$$\sum_{q=1}^{n-1} q \lg q \leq \frac{n^2}{2} \lg n - \frac{(n/2)(n/2)}{2} // \sum_{q=1}^{n-1} q \lg q \leq \frac{n^2}{2} \lg n - \frac{n^2}{8}$$

Part E

Show that the recurrence relation has solution $E[T(q)] = O(n \lg n)$

By O notation we assume $E[T(q)] \leq a \cdot q \lg q$ for some constant a and all $q < n$

From Part C

$$E[T(n)] = \frac{2}{n} \sum_{q=1}^{n-1} E[T(q)] + \Theta(n)$$

$$E[T(n)] \leq \frac{2}{n} \sum_{q=1}^{n-1} a \cdot q \lg q + \Theta(n)$$

$$E[T(n)] \leq \frac{2a}{n} \sum_{q=1}^{n-1} q \lg q + \Theta(n)$$

Using the bound from Part D

$$E[T(n)] \leq \frac{2a}{n} \left(\frac{n^2}{2} \lg n - \frac{n^2}{8} \right) + \Theta(n)$$

$$E[T(n)] \leq a \cdot n \lg n - \frac{an}{4} + \Theta(n)$$

$$E[T(n)] \leq an \lg n - \left(\frac{a}{4} - c \right) n$$

For a sufficiently large a , we get

$$E[T(n)] \leq an \lg n$$

Therefore, $E[T(n)] = O(n \lg n)$

Problem 3

27 Cards, 3 possible Colors, Shapes, and Numbers. A match is a set of cards with either all the same values, or the complete set of Colors, Shapes and Numbers

Part A

Since we are given the first 2 cards, this assumes they are already in the set of 3 cards that produces a match (either the same OR missing one element from

each set of Colors, Shapes and Numbers).

This implies that there are only 25 possible cards to choose from.

In the case that the cards have the same set of values, there is only 1 possible card in left to complete the set.

Thus $Pr(X) = 1/25$ given that $X = 1$ if the 3 cards match; 0 otherwise.

The other case is the exact same since there is only one possible card that completes the set for the values.

Thus, the possibility the card forms a match is $Pr(X) = 1/25$

Part B

We can use an Indicator Random Variable X_S , for every possible subset of 3 cards, as

$X_S = 1$ if 3 cards in subset S form a match, 0 otherwise

Thus the total number of matches is

$$X = \sum_{All\ 3-cardsubsets, S} X_S$$

We can use LOE here to get the following

$$E[X_S] = E[\sum_S X_S] = \sum_S E[X_S]$$

Since there are n cards and we choose three there are $\binom{n}{3}$ possible way to choose a subset.

$$\binom{n}{3} = n(n-1)(n-2)/3!$$

We know $n = 27$, thus $\binom{27}{3} = 27 \cdot 26 \cdot 25/6 = 2925$

Since there is only one possible choice for a match after we have chosen 2 cards we can get an expression for the number of cards that can possible complete the matching set, which is the number of expected matches.

$$Number\ Matches = 27 \cdot 26 \cdot 1/6 = 117$$

This, $P(X_S) = 117/2925 = 1/25$

Going back to finish the expectation

$$E[X_S] = \binom{n}{3} \cdot 1/25 = n(n-1)(n-2)/6 \cdot 25 = n(n-1)(n-2)/150$$

To check this, we use $n = 27$ since all cards will have been turned over.

$E = \binom{27}{3} \cdot 1/25 = 27 \cdot 26 \cdot 25/6! \cdot 25 = 17550/150 = 117//$ Thus the expression is correct.

Problem 4

Part A

The expected number of pairs in a *small* square can be calculated using an Indicator Random Variable.

There are x^2 small squares with area $1/x^2$ with n points uniformly distributed. For each pair $[i, j]$ we can use define the Indicator Random Variable $X_{ij} = 1$ if $[i, j]$ are in the same small square.

We need the expected value which we can define,

$$E[Pairs\ in\ Same\ Square] = \binom{n}{2} \cdot 1/x^2$$

We use $\binom{n}{2}$ as the ways to pick 2 points from n total points.

$$E[\text{Pairs in Same Square}] = \binom{n}{2} \cdot 1/x^2 = n(n-1)/2! \cdot x^2$$

The problem asks to find value of $x \geq 1/\sqrt{n(n-1)/2!} \cdot x^2 \geq 1$

$$n(n-1)/2 \geq x^2$$

$$\sqrt{n(n-1)/2} \geq x$$

Part B

Using Markov's Inequality, For non-negative X: $P(A \geq a) \leq E[X]/a$

Let X = number of pairs in the same small square

$$P(X \geq 1) \leq n(n-1)/2x^2$$

The problem wants to find the value less than 1

$$n(n-1)/2x^2 < 1$$

This is the same from part A, except the inequality, thus

$$\sqrt{n(n-1)/2} < x$$

This implies that the probability is bound by 1.

Part C

The max distance between 2 points in the same small square can be given by using the Pythagorean Theorem to get the diagonal length.

$$\text{Max Len} = \sqrt{2 \cdot (1/x)^2} = \sqrt{2}/x$$

Part D

Since the bounds for the upper (in part A) and the lower (in part B) are the same, Asymptotically, the bound is $\Theta(\sqrt{n(n-1)/2})$

This can be reduced to be $\Theta(1/n)$